PROBLEM 05

Plane and Glider

Original statement · Problems.pdf, p. 7

Aerospace Towed Flight

An airplane A is flying horizontally with a constant speed v_A, and is towing a glider B. The glider is gaining altitude. The tow cable has a length r, and the angle θ formed between the tow cable and the horizontal is increasing at a constant rate of $\dot{\theta}$. We want to determine the magnitude of the velocity, v_B^0, and the acceleration, a_B^0, of the glider as a function of θ and its derivatives.

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a tow plane A flies horizontally at constant speed v_A and tows a glider B that gains altitude. The tow cable has length r and forms an angle θ with the horizontal.
- **Hypotheses:** v_A and the cable length r are constant; $\mathbf{a}_A = \mathbf{0}$. The statement says that $\dot{\theta}$ is constant; the answer is requested as a function of θ and its derivatives, so we keep $\theta(t)$ general and particularise at the end ($\ddot{\theta} = 0$).
- **Constraints and degrees of freedom:** we place the origin at the altitude of A, so $y_A = 0$. The taut cable of constant length keeps B at the distance r from A: $(x_B - x_{A0} - v_A t)^2 + y_B^2 = r^2$. This is a holonomic constraint that depends explicitly on time (rheonomic) because the position of A is prescribed, and it is bilateral while the cable is taut. B then has $\mathrm{CDOF} = 2 - 1 = 1$: the angle θ.
- **Coordinates:** $\theta(t)$ for the glider relative to A; the position of A along the flight direction is $x_A(t) = x_{A0} + v_A t$.
- **Parameters and units:** v_A [m/s], r [m], θ [rad], $\dot{\theta}$ [rad/s], $\ddot{\theta}$ [rad/s^2].

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0\}$ with $\mathbf{i}_0$ horizontal in the direction of flight of A and $\mathbf{j}_0$ vertical, upward.
- **Relative position of B with respect to A:** the glider is behind and above the plane, so the vector from A to B points at the angle θ above the negative horizontal. Its components in $\mathcal{B}_0$ follow from the geometry of the cable:

$$\mathbf{r}_{B/A} = -r\cos\theta\,\mathbf{i}_0 + r\sin\theta\,\mathbf{j}_0$$

This is the only reading compatible with the glider gaining altitude as θ increases ($r\sin\theta$ grows with θ for $0 < \theta < \pi/2$).

Derivation

3.1 Kinematics

Position of the glider

> **Pedagogical Justification:** the position of B is the position of A plus the relative position, all written in the fixed basis, so that no derivative of a unit vector is needed.

$$\mathbf{r}_A = (x_{A0} + v_A t)\,\mathbf{i}_0, \qquad \mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} = (x_{A0} + v_A t - r\cos\theta)\,\mathbf{i}_0 + r\sin\theta\,\mathbf{j}_0$$

Velocity of the glider

> **Pedagogical Justification:** differentiate $\mathbf{r}_B$ in $\mathcal{S}_0$ with the chain rule, since θ depends on time and r is constant:

$$\frac{d}{dt}\left(-r\cos\theta\right) = r\sin\theta\,\dot{\theta}, \qquad \frac{d}{dt}\left(r\sin\theta\right) = r\cos\theta\,\dot{\theta}$$

$$\mathbf{v}_B^0 = \left(v_A + r\dot\theta\sin\theta\right)\mathbf{i}_0 + r\dot\theta\cos\theta\,\mathbf{j}_0, \qquad v_B^0 = \sqrt{\left(v_A + r\dot\theta\sin\theta\right)^2 + \left(r\dot\theta\cos\theta\right)^2}$$

Acceleration of the glider

Pedagogical Justification: differentiate each component of $\mathbf{v}_B^0$ with the product and chain rules:

$$\frac{d}{dt}\left(v_A + r\dot\theta\sin\theta\right) = r\ddot\theta\sin\theta + r\dot\theta^2\cos\theta, \qquad \frac{d}{dt}\left(r\dot\theta\cos\theta\right) = r\ddot\theta\cos\theta - r\dot\theta^2\sin\theta$$

$$\mathbf{a}_B^0 = \left(r\ddot\theta\sin\theta + r\dot\theta^2\cos\theta\right)\mathbf{i}_0 + \left(r\ddot\theta\cos\theta - r\dot\theta^2\sin\theta\right)\mathbf{j}_0$$

- **Magnitude.** Expanding both squares,

$$\left(r\ddot\theta\sin\theta + r\dot\theta^2\cos\theta\right)^2 = r^2\ddot\theta^2\sin^2\theta + 2r^2\ddot\theta\dot\theta^2\sin\theta\cos\theta + r^2\dot\theta^4\cos^2\theta$$

$$\left(r\ddot\theta\cos\theta - r\dot\theta^2\sin\theta\right)^2 = r^2\ddot\theta^2\cos^2\theta - 2r^2\ddot\theta\dot\theta^2\sin\theta\cos\theta + r^2\dot\theta^4\sin^2\theta$$

and adding them, the cross terms $\pm 2r^2\ddot\theta\dot\theta^2\sin\theta\cos\theta$ cancel:

$$\left(a_B^0\right)^2 = r^2\ddot\theta^2\left(\sin^2\theta + \cos^2\theta\right) + r^2\dot\theta^4\left(\cos^2\theta + \sin^2\theta\right) = r^2\ddot\theta^2 + r^2\dot\theta^4$$

3.2 Forces

Not applicable: kinematics-only problem. The motion of A and the cable geometry are given.

3.3 Equations of Motion

Not applicable: kinematics-only problem. No dynamic unknown appears.

3.4 Integration and Answers

Pedagogical Justification: the answers requested are the magnitudes of the velocity and of the acceleration of B in S_0 as functions of θ and its derivatives:

$$\boxed{v_B^0 = \sqrt{\left(v_A + r\dot{\theta}\sin\theta\right)^2 + \left(r\dot{\theta}\cos\theta\right)^2}}, \qquad \boxed{a_B^0 = \sqrt{r^2\ddot{\theta}^2 + r^2\dot{\theta}^4}}$$

- **Expanded speed.** Expanding the square in v_B^0 (valid for any $\theta(t)$, using $\sin^2\theta + \cos^2\theta = 1$): $v_B^0 = \sqrt{v_A^2 + 2v_A r\dot{\theta}\sin\theta + r^2\dot{\theta}^2}$.
- **Constant rate of the statement.** With $\dot{\theta}$ constant, $\ddot{\theta} = 0$ and $a_B^0 = r\dot{\theta}^2$.

Interpretation & Checks

- **Decoupling from the cruise speed:** the acceleration does not depend on v_A, because $\mathbf{a}_A = \mathbf{0}$ (uniform rectilinear translation). The glider accelerates only through its circular motion relative to A.
- **Limiting case $\ddot{\theta} = 0$:** the acceleration is purely centripetal, $a_B^0 = r\dot{\theta}^2$, directed along the cable toward A (circular motion of B around A).
- **Limit $\dot{\theta} \equiv 0$ (identically, so $\ddot{\theta} = 0$ too):** $\mathbf{v}_B^0 = v_A\,\mathbf{i}_0$ and $\mathbf{a}_B^0 = \mathbf{0}$: the glider moves rigidly with the plane. (If $\dot{\theta}$ only vanishes at one instant, $a_B^0 = r|\ddot{\theta}|$ there, not zero.)
- **Dimensional check:** $[r\ddot{\theta}] = [r\dot{\theta}^2] = \mathrm{m}\cdot\mathrm{s}^{-2}$ and $[v_A + r\dot{\theta}\sin\theta] = \mathrm{m/s}$, so a_B^0 is in $\mathrm{m/s^2}$ and v_B^0 in m/s.

- **Check against the official key:** the key of `Problems.pdf` is $v_B^0 = \sqrt{(v_A + r\dot\theta\sin\theta)^2 + (r\dot\theta\cos\theta)^2}$ and $a_B^0 = \sqrt{r^2\ddot\theta^2 + r^2\dot\theta^4}$ (read on the rendered page), identical to the result above. Sympy confirms both squares for an arbitrary function $\theta(t)$ with zero residual.

Source: Problems.pdf